Semantic Distance for Presburger Formula Generation

A weighted-Jaccard geometric distance for evaluating and training NL-to-formula generation

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Outline

Why surface metrics fail for formula generation

- String similarity measures notation; formula generation needs semantic equivalence.
- One character can change the defined solution set completely.
- Widely different strings can be logically equivalent.

Motivating example

Statement: Alice has at least 6 apples, at least as many as Bob's oranges.

Reference:

$$x \geq 6 \wedge x \geq y$$

Prediction:

$$x \geq 6 \wedge x > y$$

Two incompatible verdicts

- String similarity: ≈ 1 (one character changed: >= vs >).
- Geometric distance: $d_\beta > 0$ — prediction is a *strict subset*, excluding the boundary $x = y$.

Both surface similarity and semantic distance are needed to evaluate faithfully.

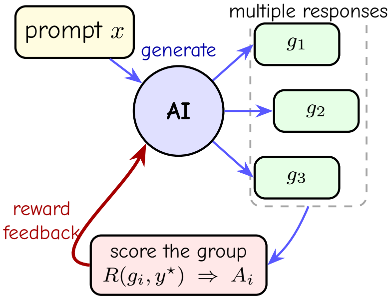
What is GRPO?

Group Relative Policy Optimization (GRPO) trains a policy from relative rewards, without a separate value model.[1] Key requirement: a reward that ranks imperfect formulas, not only exact matches.

1. Sample a group $g_1, \ldots, g_G$ for one prompt x.
2. Score each output: $R_i = R(g_i, y^\star)$.
3. Compare each reward with its group:

$$A_i = \frac{R_i - \mathrm{mean}(R_1, \ldots, R_G)}{\mathrm{std}(R_1, \ldots, R_G)}.$$

4. Reinforce $A_i > 0$ outputs; suppress $A_i < 0$ outputs.



[1] Shao et al., “DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models”, 2024.

A metric gives GRPO a direction

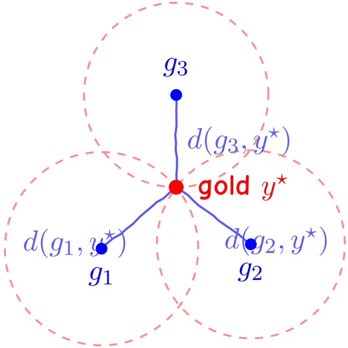
GRPO ranks sampled formulas against the supplied gold target $y^\star$:

$$R_i \uparrow \quad \text{as} \quad d(g_i, y^\star) \downarrow .$$

- Ranks imperfect outputs.
- Zero means semantic agreement.
- One shared target.

Why metricity matters

A metric gives GRPO consistent semantic geometry.



2D: three non-collinear anchors localize the supplied target. General metric spaces require a resolving set.

Contributions

1. A weighted-Jaccard metric for Presburger-definable sets.
2. A computable pipeline: QE $\rightarrow$ polyhedra $\rightarrow$ generating functions.
3. String-only GRPO $\rightarrow$ distance-aware GRPO: parsability $76.5\% \rightarrow 98.6\%$; adjusted distance $0.295 \rightarrow 0.167$.

Presburger arithmetic

Definition (Presburger arithmetic)

The first-order theory of $(\mathbf{Z}; +, \leq)$ with constants, Boolean connectives, and quantifiers.[2]
No multiplication between variables. Atomic constraints:

$$a_1 x_1 + \cdots + a_d x_d \leq b, \qquad a_1 x_1 + \cdots + a_d x_d = b, \qquad a_i, b \in \mathbf{Z}.$$

Composite formulas via $\wedge, \vee, \neg, \forall, \exists$.

- Decidable with quantifier elimination.
- Definable sets are periodic polyhedral lattice sets.

[2] Cooper, "Theorem Proving in Arithmetic without Multiplication", 1972; Ginsburg and Spanier, "Semigroups, Presburger Formulas, and Languages", 1966.

Quantifier elimination for Presburger arithmetic

Theorem (QE for Presburger arithmetic)
For any Presburger $\varphi(\mathbf{x})$ there is a quantifier-free $\psi(\mathbf{x})$ defining the same subset of $\mathbf{Z}^d$. ψ is a finite Boolean combination of linear inequalities, linear equalities, and ***congruence conditions***

$$a_1x_1 + \cdots + a_dx_d \equiv c \pmod{m}.$$

QE turns logic into finitely many linear regions with periodic constraints.

Example: eliminating a quantifier

Example (Quantifier elimination)

$$\exists z \, (x = 2z + 1 \land 0 \le z \land z \le 2)$$

$0 \le z \le 2$ bounds z; $x = 2z + 1$ links x to z. Eliminating z:

$$x = 1 \lor x = 3 \lor x = 5 \quad \Longleftrightarrow \quad 1 \le x \le 5 \land x \equiv 1 \pmod 2.$$

General pattern

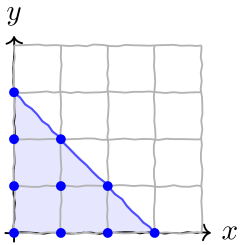
Bounds yield thresholds; linear dependencies yield congruences.

Formulas as lattice-point sets

$$S_\varphi = \{\mathbf{x} \in \mathbf{Z}^d \mid \varphi(\mathbf{x})\}, \qquad \wedge \leftrightarrow \cap, \qquad \vee \leftrightarrow \cup, \qquad \neg \leftrightarrow \text{complement.}$$

Example (Basic Presburger formulas)

- $x + y \leq 3 \wedge x \geq 0 \wedge y \geq 0$ — lattice points of a 2D simplex.
- $\exists z\, (x = 2z)$ — the even integers.



Weighted Presburger measure

Definition (Weighted Presburger measure)

Fix $\beta > 0$, $\varphi(\mathbf{x})$ over $\mathbf{x} \in \mathbf{Z}^d$:

$$\mu_\beta(\varphi) = \frac{1}{Z_{\beta,d}} \sum_{\mathbf{x} \in S_\varphi} e^{-\beta \|\mathbf{x}\|_1}, \qquad Z_{\beta,d} = \left(\frac{1 + e^{-\beta}}{1 - e^{-\beta}} \right)^d .$$

Theorem (Well-definedness and boundedness)

For any Presburger φ and $\beta > 0$: $\mu_\beta(\varphi)$ is well-defined and $0 \leq \mu_\beta(\varphi) \leq 1$.

- positive weights $\Rightarrow \mu_\beta \geq 0$;
- $S_\varphi \subseteq \mathbf{Z}^d \Rightarrow$ numerator $\leq Z_{\beta,d}$;
- therefore $\mu_\beta \in [0, 1]$.

μ_β is an exponentially local prior on lattice points.

Computing the measure: orthant decomposition

QE $\Rightarrow$ quantifier-free formula $\Rightarrow$ finite disjoint union of rational polyhedra:

$$S_\varphi = \bigsqcup_{k=1}^{m} (P_k \cap \mathbf{Z}^d).$$

Split into 2^d orthants; sign flips make $\|\mathbf{x}\|_1$ linear.

$$\mu_\beta(\varphi) = \frac{1}{Z_{\beta,d}} \sum_{k=1}^{m} \sum_{\mathbf{x} \in P_k \cap \mathbf{Z}^d} e^{-\beta\|\mathbf{x}\|_1}.$$

Cost: 2^d orthants grow prohibitive as d increases.

Example: one-dimensional threshold formulas

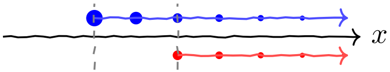
Example (1D threshold formulas)

$d = 1$: $p(x) \equiv x \geq 0$, $q(x) \equiv x \geq 2$. $S_p = \{0, 1, 2, \ldots\}$, $S_q = \{2, 3, 4, \ldots\}$.
$Z_{\beta,1} = \frac{1+e^{-\beta}}{1-e^{-\beta}}$,

$$\mu_\beta(p) = \frac{\sum_{n\geq 0} e^{-\beta n}}{Z_{\beta,1}}, \qquad \mu_\beta(q) = \frac{\sum_{n\geq 2} e^{-\beta n}}{Z_{\beta,1}}.$$

$p \wedge q \equiv q$, $p \vee q \equiv p \Rightarrow$

$$d_\beta(p,q) = 1 - e^{-2\beta}.$$



At $\beta = 1$, unmatched points 0 and 1 dominate.

Generating functions and Barvinok's algorithm

Map into nonnegative orthant, then use **Ehrhart / generating-function** theory:[3]

$$Q \subseteq \mathbf{R}^d_{\geq 0}, \qquad G_Q(\mathbf{t}) = \sum_{\mathbf{y} \in Q \cap \mathbf{Z}^d} t_1^{y_1} \cdots t_d^{y_d}.$$

- Barvinok: rational encoding in polynomial time for fixed d.
- Evaluate $t_i = e^{-\beta}$.
- Compute decomposed polyhedra with **LattE**.

[3] Woods, "The Unreasonable Ubiquitousness of Quasi-polynomials", 2014; A. I. Barvinok, "A Polynomial Time Algorithm for Counting Integral Points in Polyhedra When the Dimension is Fixed", 1994; A. Barvinok, "Integer Points in Polyhedra", 2008; De Loera et al., "Effective lattice point counting in rational convex polytopes", 2004.

Order relations and additivity

Every weight is strictly positive $\Rightarrow$:

$$A \subseteq B \;\Rightarrow\; \mu_\beta(A) \le \mu_\beta(B) \qquad \text{(monotonicity)}$$

$$A \subseteq B \cup C \;\Rightarrow\; \mu_\beta(A) \le \mu_\beta(B) + \mu_\beta(C) \qquad \text{(subadditivity)}$$

These underpin the triangle inequality for

$$\delta_\beta(p, q) = \mu_\beta(p \triangle q), \qquad p \triangle q = (p \wedge \neg q) \vee (q \wedge \neg p).$$

The weighted Jaccard distance

$$d_\beta(p,q) = 1 - \frac{\mu_\beta(p \wedge q)}{\mu_\beta(p \vee q)} = \frac{\mu_\beta(p \triangle q)}{\mu_\beta(p \vee q)},$$

with $d_\beta(p,q) = 0$ when $\mu_\beta(p \vee q) = 0$ (both empty).

- Discrete weighted analogue of the classical **Jaccard distance**.[4]
- β controls locality: large β concentrates near the origin; small β weights distant points more.

[4] Jaccard, "Étude comparative de la distribution florale dans une portion des Alpes et du Jura", 1901.

Lemma: basic identities for μ_β

Lemma (Basic identities)
For Presburger p, q:

1. $\mu_\beta(p) \geq 0$.
2. $\mu_\beta(p \vee q) = \mu_\beta(p) + \mu_\beta(q) - \mu_\beta(p \wedge q)$.
3. $\mu_\beta(p \triangle q) = \mu_\beta(p \vee q) - \mu_\beta(p \wedge q)$.

- Nonnegative weights $\Rightarrow \mu_\beta(p) \geq 0$.
- Inclusion–exclusion removes the double-counted $p \wedge q$.
- Union minus overlap gives $p \triangle q$.

Lemma: positivity & symmetric-difference form

Lemma (Positivity and symmetric-difference representation)
If $\mu_\beta(p \vee q) > 0$*:*

$$d_\beta(p,q) = \frac{\mu_\beta(p \triangle q)}{\mu_\beta(p \vee q)}.$$

Moreover, if p, q *define different subsets of* $\mathbf{Z}^d$*, then* $d_\beta(p,q) > 0$*.*

- The previous identities give the displayed fraction.
- $p \neq q \Rightarrow p \triangle q$ contains $\mathbf{a}$.
- Strictly positive weight at $\mathbf{a} \Rightarrow \mu_\beta(p \triangle q) > 0 \Rightarrow d_\beta(p,q) > 0$.

Theorem: metricity

Theorem (Metricity of the Presburger distance)

*Fix $\beta > 0$. Then d_β is a **metric** on the family of Presburger-definable subsets of $\mathbf{Z}^d$.*

- Definition $\Rightarrow$ non-negativity and symmetry.
- Finite measure is monotone and modular $\Rightarrow$ weighted-Jaccard triangle inequality; $A \triangle C \subseteq (A \triangle B) \cup (B \triangle C)$ is one ingredient.
- Strictly positive point weights $\Rightarrow \mu_\beta(A \triangle B) = 0 \Rightarrow A = B$ (identity of indiscernibles).

Experimental setup

Task NL $\rightarrow$ Presburger formula
Model `Qwen/Qwen3-4B`.[5]
Data `train/dev/test` JSONL; 2,000 test examples
Pipeline SFT $\rightarrow$ LoRA merge $\rightarrow$ GRPO

Same GRPO budget:
100 steps · batch 32 · 16 samples/prompt · lr 10^{-5}
temp 0.8 · max length 128 · $\beta = 1$

Rewards:

- `string_similarity`
- `presburger_distance`

`https://github.com/tani/presburger-distance-naloma2026`

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[5] Yang et al., "Qwen3 Technical Report", 2025; Shao et al., "DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models", 2024.

Data format and generation

JSONL item: natural-language problem $\rightarrow$ target formula.

```
{"uuid7": "...",
 "problem": "...",
 "formula": "(A x)(E y)[-x + 5 z <= 7 <==> 2 y - 3 z = -6]"}
```

- Template walks generate formulas.
- An LM verbalizes each formula.

Reward functions

$$R_{\mathsf{sim}}(g, r) = \mathrm{SeqMatch}(g, r)$$

$$R_{\mathsf{dist}}(g, r) = \begin{cases} -1.0 & \text{invalid output} \\ 0.5\,(1 - D(g, r)) + 0.5\ \mathrm{SeqMatch}(g, r) & \text{otherwise} \end{cases}$$

Distance timeout $\Rightarrow$ distance component set to 0.0. **String similarity shapes syntax early; semantic distance ranks valid formulas.**

$$D(p, q) = 1 - \frac{\mu(p \wedge q)}{\mu(p \vee q)}, \qquad D_{\mathsf{adj}}(g, r) = \begin{cases} D(g, r) & \text{computable} \\ 1.0 & \text{otherwise} \end{cases}$$

Invalid or timed-out outputs receive worst-case adjusted distance.

Main results

Run	Model	Parsable	Exact/Equiv	String Sim	Raw Dist	Adj. Dist
string_similarity	Base	9.90%	0.50%	0.3814	0.2115	0.9223
string_similarity	SFT	87.70%	68.45%	0.9561	0.0533	0.1778
string_similarity	GRPO	76.50%	47.90%	0.9410	0.0740	0.2953
presburger_distance	Base	9.90%	0.50%	0.3814	0.2111	0.9223
presburger_distance	SFT	87.45%	67.30%	0.9544	0.0555	0.1821
presburger_distance	GRPO	98.55%	54.05%	0.9382	0.1462	0.1667

Distance-aware GRPO reaches 98.55% parsability and the best adjusted distance (0.1667).

Head-to-head comparison

2,000 shared test cases:

Comparison	Count
both exact	890
only string_similarity exact	68
only presburger_distance exact	191
only string_similarity parsable	7
only presburger_distance parsable	448
string_similarity better (adj. dist)	62
presburger_distance better (adj. dist)	474
tied (adj. dist)	1464

Distance-aware GRPO is uniquely parsable on 448 cases and has lower adjusted distance on 474.

Error analysis: quantifier handling is the bottleneck

The rewards trade malformed quantifiers for dropped quantifiers.

Pattern	sim	dist
invalid bracketed quantifier after quantified ref.	336	0
all quantifiers dropped after quantified ref.	158	597
other invalid output after quantified ref.	117	14
invalid GRPO output	470	29

- String reward: keeps quantifier tokens but often breaks syntax.
- Distance reward: restores syntax but drops prefixes in 597/611 cases.

Representative example: syntax repair

Reference: `[3 x - 4 y - 5 z >= 9 \/ 9 x + 5 y - 5 z /= -7]`

SFT output unparsable

```
(3 x - 4 y - 5 z >= 9 \/ 9 x + 5 y - 5 z /= -7)
```

GRPO + `distance_penalized` exact

```
[3 x - 4 y - 5 z >= 9 \/ 9 x + 5 y - 5 z /= -7]
```

Opposite failure

Reference: `(A x)(E y)[-x + 5 z <= 7 <==> 2 y - 3 z = -6]`

similarity_only

```
[A x][E y][5 z - x <= 7 <==> 2 y - 3 z = -6]
```

distance_penalized

```
[5 z - x <= 7 <==> 2 y - 3 z = -6]
```

- String reward: unparsable quantifier brackets.
- Distance reward: parsable, but quantifiers disappear.

Case studies: syntax vs. semantics diverge

Case A — reference `-9 x + 2 z <= -4`:

- `similarity_only`: `[-9 x + 2 z <= -4]`
  semantically equivalent; not exact
- `distance_penalized`: `-9 x + 2 z <= -4` exact

Case B — reference `[-10 x - 5 y = -5 <==> -9 x - 9 y >= -5]`:

- `distance_penalized`: `[-10 x + (-5 y) = -5 <==> -9 x + (-9 y) >= -5]`
  unparsable: parenthesized negatives rejected

Exact-string, syntactic, and semantic correctness can diverge.

Scope in the NLP landscape

- Controlled formal-language case study: Presburger arithmetic excludes nonlinear operations and higher-order constructs.
- QE and polyhedral counting make this restricted setting tractable.
- The metric is a foundation, not a universal semantic measure.

Synthetic data: a deliberate choice

- Formula templates provide controlled semantic targets.
- Open-domain noise would confound reward attribution.
- Results remain conditional on a pretrained mathematical LM.

Computational complexity and scalability

- Distance reward improves parsability and adjusted distance; string-only reward degrades syntactic validity.
- QE plus LattE scales poorly with dimension and coefficient size, echoing integer-programming hardness.[6]
- Harder formulas remain an open scaling test.

[6] Papadimitriou, "On the complexity of integer programming", 1981.

Limitations

- **Scale:** simple formulas; larger dimensions, coefficients, and quantifier depth remain untested; QE/counting may bottleneck training.
- **Generality:** one formal-translation task; transfer and pretrained-model dependence are unresolved.
- **Reward design:** quantifier prefixes remain fragile; no pure-distance GRPO ablation.

Conclusion

- **Best GRPO signal:** distance-aware reward improves parsability and adjusted semantic distance.
- **Best exact reproduction:** SFT remains strongest.
- **Next step:** reward quantifier structure explicitly.

Take-home
A computable semantic metric guides GRPO beyond string overlap—but structure needs its own signal.

Thank you.

https://github.com/tani/presburger-distance-naloma2026